

David Oluwatimilehin “Timi” Folayan

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EDUCATION

ROSE-HULMAN INSTITUTE OF TECHNOLOGY

Bachelor of Science

Terre Haute, IN

Expected 2027

Major in Software Engineering; Minor in Cybersecurity

Relevant Coursework: Object-Oriented Development, Operating Systems, Web Development, DSA, Digital Systems, Databases, Diff Equations, Practical Security, Mathematical Proofs, Formal Methods, Combinatorics

SKILLS

Languages and Technologies: Java, SQL, C#, JavaScript, HTML/CSS, C, C++, Python, Verilog, React, Mathematica

Development Skills and Tools: Git, VS Code, Eclipse IDE, Unit Testing, Firebase, Quartus, ModelSim, Arduino, IntelliJ, MSS

WORK EXPERIENCE

RHV | Gas Valve Calibration System

Terre Haute, IN

Software Engineer Intern

Summer 2025

- Sole software engineer on a multidisciplinary team with electrical, computer, and mechanical engineers, leading the design and development of the system's full software stack
- Built a responsive, multithreaded WinForms application in C# that served as the central interface connecting mechanical components and electrical systems
- Developed real-time UI to monitor valve calibration status, regulate test sequences, and visualize system diagnostics across six simultaneous test units
- Implemented hardware communication using Modbus TCP/IP and serial protocols to control PLCs and programmable power supplies
- Integrated electrical inputs and mechanical feedback through software, allowing for seamless coordination of thermal, pressure, and flow testing operations
- Supported the successful calibration of over 1,000,000 gas valves, enabling high-throughput production with minimal operator intervention

RHIT

Terre Haute, IN

Teacher Assistant

Spring 2025 – Present

- Supported undergraduate students in maturing formal method ideas, including logic, specifications, and program correctness
- Assisted with grading assignments and providing detailed feedback on formal proofs and model-based designs

Ask Rose

Terre Haute, IN

Learning Advisor

Winter 2024 – Present

- Assisted students from grades 6-12 with STEM homework problems, focusing on subjects such as mathematics, physics, and computer science, resulting in a 99% student satisfaction rate
- Provided clear explanations and guided students through problem-solving techniques to develop a deeper understanding of core concepts, leading to a measurable improvement in students' skills, with 80% reporting increased confidence

PROJECTS

Lost and Found Database | SQL, Java, Eclipse IDE, Microsoft SQL Server

Winter 2024

- Designed and developed a comprehensive database system for a lost and found application using Microsoft SQL Server, reducing item retrieval time by 40%
- Created an intuitive Java-based user interface to manage and track lost and found items, improving ease of use for admins and users
- Applied object-oriented design principles and integrated Java-SQL connectivity for seamless database interaction, enhancing system maintainability and scalability

LEADERSHIP

African Student Union, Vice President

2024 – Present

National Society of Black Engineers, Public Relations Chair

2024 – Present

Linux Users Group, Computer Security Club, Member

2023 – Present